

```
# Import Required Libraries
```

```
import pandas as pd
```

```
import pickle
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.model_selection import cross_val_score
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
# -----
```

```
# Load Dataset
```

```
# -----
```

```
df = pd.read_csv("loan_data.csv")
```

```
# -----
```

```
# Remove Loan_ID
```

```
# -----
```

```
if "Loan_ID" in df.columns:
```

```
    df.drop("Loan_ID", axis=1, inplace=True)
```

```
# -----
```

```
# Handle Missing Numerical Values
```

```
# -----
```

```
numerical_columns = [
```

```
    "ApplicantIncome",
```

```
    "CoapplicantIncome",
```

```
    "LoanAmount",
```

```
    "Loan_Amount_Term",
```

```
    "Credit_History"
```

```
]
```

```
for column in numerical_columns:
```

```
    df[column] = df[column].fillna(df[column].mean())
```

```
# -----
```

```
# Handle Missing Categorical Values
```

```
# -----
```

```
categorical_columns = [
```

```
    "Gender",
```

```
    "Married",
```

```
    "Dependents",
```

```
    "Education",
```

```
    "Self_Employed",
```

```
    "Property_Area",
```

```

    "Loan_Status"
]

for column in categorical_columns:
    df[column] = df[column].fillna(df[column].mode()[0])

# -----
# Encode Categorical Features
# -----

encoder = LabelEncoder()

for column in categorical_columns:
    df[column] = encoder.fit_transform(df[column].astype(str))

# -----
# Separate Features and Target
# -----

X = df.drop("Loan_Status", axis=1)
y = df["Loan_Status"]

# -----
# Split Dataset
# -----

```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.20,  
    random_state=42,  
    stratify=y  
)
```

```
# -----  
# Train Random Forest Model  
# -----
```

```
model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)
```

```
model.fit(X_train, y_train)
```

```
# -----  
# Cross Validation  
# -----
```

```
scores = cross_val_score(  
    model,  
    X,
```

```
y,  
cv=5  
)  
  
print("=" * 50)  
print("Cross Validation Scores")  
print("=" * 50)  
  
print(scores)  
  
print("\nAverage Accuracy :", scores.mean())  
  
# -----  
# Save Model  
# -----  
  
filename = "loan_prediction_model.pkl"  
  
with open(filename, "wb") as file:  
    pickle.dump(model, file)  
  
print("\nModel saved successfully as", filename)
```